How often a curve over F_{p^6} lands in a Joux–Vitse weak class

Protocol frozen before the runs, two disclosed amendments. Every number is an unverified assertion [pcc:unverified].

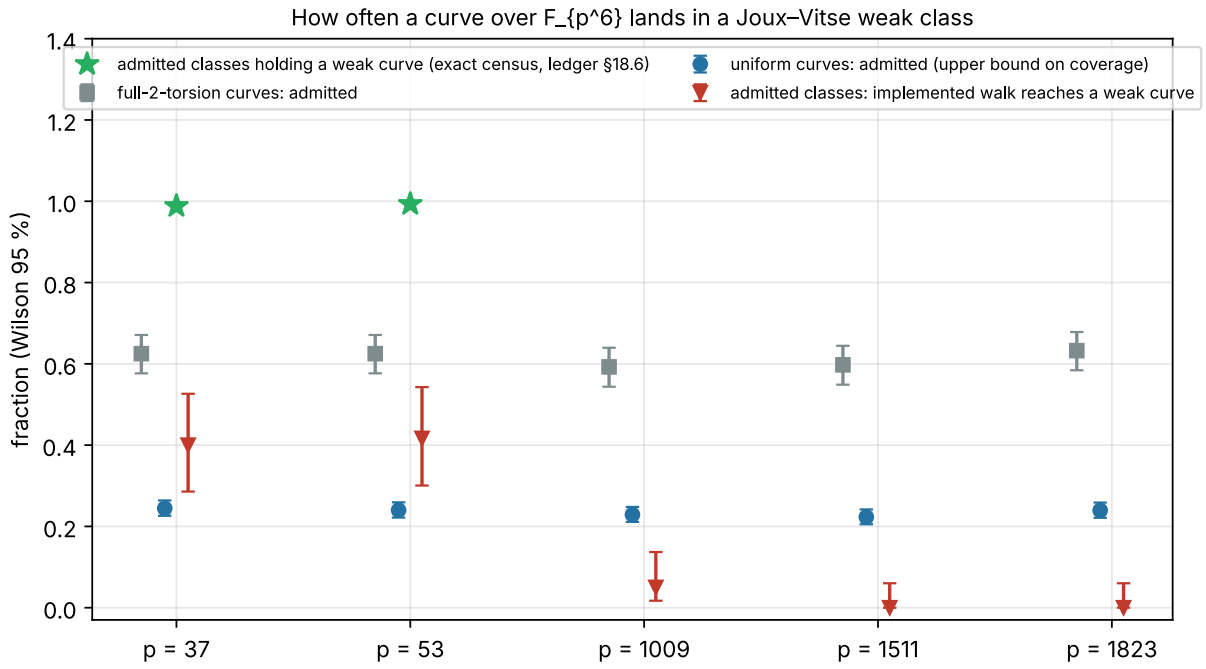

Figure 1: Admitted shares, census truth at small p, and the reach of the implemented walk.

- **Admission:** 22–24.5 % of uniform curves pass the proved necessary condition at every p from 37 to 1823. Among curves with full 2-torsion it is 59–63 %. This is the upper bound on coverage.
- **Census:** at small p, 98.8–99.6 % of admitted classes hold a weak curve. At p = 1009, three random admitted classes were reached: the first large-field witnesses.
- **Implemented walk:** it reaches about 40 % of admitted classes at p = 37–53, 3 of 60 at p = 1009, and 0 of 60 at p = 1511 and 1823. Its odd jumps need rational ℓ-torsion points.
- **Operational coverage:** about 10 % of curves at toy sizes, about 1 % at p = 1009.
- **Cost:** successful random-start walks cost 0.008–0.14 × rho at p = 1009.
- **Consequence:** the walk charge holds only for starts the walk can reach. A full class-group walk using kernel polynomials of degree $(\ell-1)/2$ is the next construction step.